ARJUNA JEE 2027

Lecture -04 | Mathematics

BINOMIAL THEOREM





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Topics *to be covered*

A **Numerically Greatest Term**

B **Properties of Binomial Coefficients**

C **Question Practice**

D



H.W. Discussion

DIBY-05

Find the greatest value of the term independent of x in the expansion of $\left(x\sin\alpha + \frac{1}{x}\cos\alpha\right)^{10}$, where $\alpha \in \mathbb{R}$.

(A) $\frac{63}{8}$ (B) $\frac{63}{16}$ (C) $\frac{63}{32}$ (D) $\frac{63}{64}$

[Ans. : (A)]

$$\left(x\sin\alpha + \frac{1}{x}\cos\alpha\right)^{10} \rightarrow T_{r+1} = {}^{10}C_r \cdot (x\sin\alpha)^{10-r} \cdot \left(\frac{\cos\alpha}{x}\right)^r$$

$$= {}^{10}C_r \cdot (\sin\alpha)^{10-r} \cdot (\cos\alpha)^r \cdot x^{10-2r} \quad (10-2r=0 \Rightarrow r=5)$$

$$\rightarrow {}^{10}C_5 (\sin\alpha)^5 (\cos\alpha)^5 = {}^{10}C_5 \frac{(\overbrace{2\sin\alpha\cos\alpha}^{1})^5}{2^5}$$

Question [JEE Mains-2024]

Number of integral terms in the expansion of $\left\{7^{\left(\frac{1}{2}\right)}+11^{\left(\frac{1}{6}\right)}\right\}^{824}$ is equal to _ _ _ _ _ _ _ _

Ans: 138

Done!!

Question [JEE Mains-2024]

If the constant term is expansion of $\left(1+2x-3x^3\right)\left(\frac{3}{2}x^2-\frac{1}{3x}\right)^9$ is p, then $108p$ is equal to

54

$T_{r+1} = {}^9C_r \left(\frac{3}{2}x^2\right)^{9-r} \cdot \left(\frac{-1}{3x}\right)^r$

$= {}^9C_r \cdot \frac{3^{9-2r}}{2^{9-r}} (-1)^r \cdot x^{18-3r}$

$18-3r = -3$

$21 = 3r$

$7 = r$

$18-3r = 0$

$r = 6$

$18-3r = -1$ ✓

$19 = 3r \rightsquigarrow r = \frac{19}{3}$ ×

$1\,x^0 \times \left(\frac{{}^9C_6\, 3^{-3}\times 1}{2^3}\right)x^0 = \frac{{}^9C_6\, 3^{-3}}{2^3}\,x^0$

$2x^1 \times (0)\,x^{-1} = 0\,x^0$

$+3x^3 \times \left(\frac{{}^9C_7\, 3^{-5}(+1)}{2^2}\right)x^{-3} = \frac{{}^9C_7\, 3^{-5}\times 3}{2^2}\,x^0$

add

$p = \frac{1}{2}$

Question [JEE (Adv.)-2014]

Coefficient of x^{11} in the expansion of $(1+x^2)^4(1+x^3)^7(1+x^4)^{12}$ is

(1) x^0, x^2, x^4, x^6, x^8 (2) $x^0, x^3, x^6, x^9 \ldots x^{21}$ $x^0, x^4, x^8, x^{12} \ldots x^{48}$

$\square \times \square \times \square = x^{11}$

$^4C_0\, x^0 \times {}^7C_1\, x^3 \times {}^{12}C_2\, x^8 = {}^4C_0\, {}^7C_1\, {}^{12}C_2$

$^4C_2\, x^4 \times {}^7C_1\, x^3 \times {}^{12}C_1\, x^4 = {}^4C_2\, {}^7C_1\, {}^{12}C_1$

$^4C_4\, x^8 \times {}^7C_1\, x^3 \times {}^{12}C_0\, x^0 = {}^4C_4\, {}^7C_1\, {}^{12}C_0$

$^4C_1\, x^2 \times {}^7C_3\, x^9 \times {}^{12}C_0\, x^0 = {}^4C_1\, {}^7C_3\, {}^{12}C_0$

add

A. 1051 $^4C_r\, x^{2r}$

B. 1106 $^7C_r\, x^{3r}$

C. 1113 $^{12}C_r\, x^{4r}$ 84%

D. 1120

Question [JEE Mains-2019]

The coefficient of x^{18} in the product $(1+x)(1-x)^{10}(1+x+x^2)^9$ is

A 84

B −126

C −84

D 126



$(1+x)(1-x)^{10}(1+x+x^2)^9$

$(1+x)(1-x)^1 \quad (1-x)^9(1+x+x^2)^9$

$(1-x^2)(1-x^3)^9$

$x^0, x^3, x^6, \ldots$

Coeff of x^{18} →

$1x^0 \times {}^9C_6(1)x^{18} = {}^9C_6\, x^{18}$

$-1x^2 \times 0 \; x^{16} = 0\; x^{18}$

${}^9C_r(-x^3)^r = {}^9C_r(-1)^r \cdot x^{3r}$

${}^9C_6 = 84$

DIBY – (X)

• Coeff of x^{26}

Question [JEE Mains-2024]

In the expansion of $(1+x)(1-x^2)\left(1+\frac{3}{x}+\frac{3}{x^2}+\frac{1}{x^3}\right)^5$, $x \neq 0$, the sum of the coefficient of x^3 and x^{-13} is equal to _____

-1 $\quad$ 119 $\quad$ 118

coeff of x^3 in $\rightarrow$ $\dfrac{(1+x-x^2-x^3)(1+x)^{15}}{x^{15}}$

fns

coeff. of x^3 in $\rightarrow \dfrac{(1+x-x^2-x^3)(1+x)^{15}}{x^{15}} = -1$

OR

coeff. of x^{18} in $\rightarrow (1+x-x^2-x^3)(1+x)^{15}$ $\rightarrow {}^{15}C_r\, x^r$

$x^0 \quad 0\ x^{18}$

$x^1 \quad 0\ x^{17}$

$x^2 \quad 0\ x^{16}$

$-1 \cdot x^3 \times {}^{15}C_{15}\, x^{15} = (-1)\, x^{18}$

coeff. of x^{-13} in $\rightarrow \dfrac{(1+x-x^2-x^3)(1+x)^{15}}{x^{15}}$

OR

coeff. of x^2 in $\rightarrow (1+x-x^2-x^3)(1+x)^{15}$

$(1+x)^{15} \rightarrow {}^{15}C_r\, x^r \rightarrow$ r ∈ W

$1\,x^0 \times {}^{15}C_2\, x^2 = {}^{15}C_2\, x^2$ (105)

$1\,x^1 \times {}^{15}C_1\, x^1 = {}^{15}C_1\, x^2$

$-1\,x^2 \times {}^{15}C_0\, x^0 = -1\, x^2$

$-1\,x^3 \times 0\, x^{-1} = 0\, x^2$

$105 + 15 - 1 = 119$

Question [JEE Mains-2024]

If the coefficient of x^{30} in the expansion of $\left(1+\frac{1}{x}\right)^{6}\left(1+x^{2}\right)^{7}\left(1-x^{3}\right)^{8}$; $x \neq 0$ is α, then $|\alpha|$ equals



RAPID FIRE DIBY'S

DIBY-01

In the binomial expansion of $(a-b)^n, n \geq 5$, the sum of 5th and 6th terms is zero, then $\frac{a}{b}$ equals

(A) $\frac{6}{n-5}$ (B) $\frac{n-5}{6}$ (C) $\frac{n-4}{5}$ (D) $\frac{5}{n-4}$

[Ans. : (C)]

DIBY-02

Number of rational terms in the expansion of $(\sqrt{2}+\sqrt[4]{3})^{100}$ is :

(A) 25 (B) 26 (C) 27 (D) 28

[Ans. : (B)]

RAPID FIRE DIBY'S

DIBY-03

Find the coefficient of x^2 in the expansion of $(1-x+2x^2)\left(x+\frac{1}{x}\right)^{10}$

(A) 120 (B) 210 (C) 720 (D) 504

[Ans. : (C)]

DIBY-04

If a_r denotes the coefficient of x^r in the expansion of $(1-x)^{2n-1}$, then the value of $a_{r-1}+a_{2n-r}=$

(A) –1 (B) 1 (C) $2a_r$ (D) 0

[Ans. : (D)]

RAPID FIRE DIBY'S

DIBY-05

Find the term independent of x in the expansion of $\left(1+x+2x^3\right)\left(\frac{3x^2}{2}-\frac{1}{3x}\right)^9$.

[Ans. : $\frac{17}{54}$]

DIBY-06

For natural numbers m, n if $(1-y)^m(1+y)^n = 1+a_1y+a_2y^2+\cdots$, and $a_1 = a_2 = 10$, then (m, n) is–

(A) $(45, 35)$ (B) $(35, 45)$ (C) $(20, 45)$ (D) $(35, 20)$

[Ans. : (B)]

Coeff of $y^1 \rightarrow (1-y)^m (1+y)^n$

$^mC_r(-y)^r$

$^nC_r y^r$

$^mC_0\, y^0 \times {}^nC_1\, y^1 = n\, y^1$

$-{}^mC_1\, y^1 \times {}^nC_0\, y^0 = -m\, y^1$

$a_1 = n - m = 10$

#DIBY-06 Notes

For natural numbers m, n if $(1-y)^m(1+y)^n = 1 + a_1y + a_2y^2 + \cdots$, and $a_1 = a_2 = 10$, then (m, n) is–

Coeff of y^2

Coeff of y^1

(A) $(45, 35)$ **(B)** $(35, 45)$ **(C)** $(20, 45)$ **(D)** $(35, 20)$

[Ans. : (B)]

Coeff of $y^2 \rightarrow (1-y)^m (1+y)^n$

$^mC_r(-y)^r$

$^nC_r y^r$

$^mC_0\ y^0 \times {}^nC_2\ y^2 = {}^nC_2 y^2$

$-{}^mC_1\ y^1 \times {}^nC_1\ y^1 = -mn\ y^2$

$^mC_2\ y^2 \times {}^nC_0\ y^0 = {}^mC_2 y^2$

$a_2 = {}^nC_2 - mn + {}^nC_2 = 10$

Coeff. of y^2

Coeff. of y^1

DIBY-06

For natural numbers m, n if $(1-y)^m(1+y)^n = 1 + a_1y + a_2y^2 + \cdots$, and $a_1 = a_2 = 10$, then (m, n) is–

(A) $(45, 35)$ **(B)** $(35, 45)$ **(C)** $(20, 45)$ **(D)** $(35, 20)$

[Ans. : (B)]

$n - m = 10$

$n + m = 80$

$2n = 90$

$n = 45$

$n = 35$

$${}^{n}C_2 - mn + {}^{m}C_2 = 10$$

$$\frac{n(n-1)}{2} - mn + \frac{m(m-1)}{2} = 10$$

$$n^2 - n - 2mn + m^2 - m = 20$$

$$(n-m)^2 - n - m = 20$$

$100 - 20 = n + m$

RAPID FIRE DIBY'S

DIBY-07

Find the coefficient of x^m in $(1+x)^m + (1+x)^{m+1} + \cdots + (1+x)^n$, where $m \leq n$.

(A) ${}^{n+1}C_{m+1}$ (B) ${}^{n}C_{m+1}$ (C) ${}^{n+1}C_{m}$ (D) ${}^{n}C_{m}$

[Ans. : (A)]

DIBY-08

The sum of the coefficients of all the even powers of x in the expansion of $(2x^2 - 3x + 1)^{11}$ is –

(A) 2.6^{10} (B) 3.6^{10} (C) 6^{11} (D) None of these

[Ans. : (B)]

RAPID FIRE DIBY'S

DIBY-07

Find the coefficient of x^m in $(1+x)^m + (1+x)^{m+1} + \cdots + (1+x)^n$, where $m \leq n$.

(A) ${}^{n+1}C_{m+1}$ (B) ${}^{n}C_{m+1}$ (C) ${}^{n+1}C_{m}$ (D) ${}^{n}C_{m}$

[Ans. : (A)]

G.P.

$a = (1+x)^m$

$r = (1+x)$

not $= n-m+1$.

ex:-

Coeff of x^7 in $\left((1+x)^1+(1+x)^2+(1+x)^3+(1+x)^4+\ldots\ldots(1+x)^{15}\right)$

→ G.P.

$a = 1+x,\ r = 1+x,\ n = 15$

$$(1+x)\left(\frac{(1+x)^{15}-1}{x+x-1}\right)$$

Coeff of x^7 in $\rightarrow \dfrac{(1+x)^{16}-(1+x)}{x}$

Coeff of x^8 in $\rightarrow (1+x)^{16}-(1+x)$

→ ${}^{16}C_8$

NUMERICALLY GREATEST TERM

without sign

$$(a+b)^n$$

$$|T_r| \leq |T_{r+1}| \geq |T_{r+2}|$$

$$\left|{}^nC_{r-1}\, a^{n-r+1} b^{r-1}\right| \leq \left|{}^nC_r\, a^{n-r} b^r\right| \geq \left|{}^nC_{r+1}\, a^{n-r-1} b^{r+1}\right|$$

$$(2+3)^4 = \underset{1}{{}^4C_0}\,2^4 + \underset{4}{{}^4C_1}\,2^3 3^1 + \underset{6}{{}^4C_2}\,2^2 3^2 + \underset{4}{{}^4C_3}\,2^1 \cdot 3^3 + \underset{1}{{}^4C_4}\,3^4$$

$$= \underbrace{16}_{{}^4C_0 2^4} + \underbrace{96}_{{}^4C_1 2^3 3^1} + \underbrace{216}_{{}^4C_2 2^2 3^2} + \underbrace{216}_{{}^4C_3 2^1 \cdot 3^3} + \underbrace{81}_{{}^4C_4 3^4}$$

$$(2-3)^4 = \underset{1}{{}^4C_0}\,2^4 - \underset{4}{{}^4C_1}\,2^3 3^1 + \underset{6}{{}^4C_2}\,2^2 3^2 - \underset{4}{{}^4C_3}\,2^1 \cdot 3^3 + \underset{1}{{}^4C_4}\,3^4$$

$$= \underbrace{16}_{{}^4C_0 2^4} - \underbrace{96}_{{}^4C_1 2^3 3^1} + \underbrace{216}_{{}^4C_2 2^2 3^2} - \underbrace{216}_{{}^4C_3 2^1 \cdot 3^3} + \underbrace{81}_{{}^4C_4 3^4}$$



NUMERICALLY GREATEST TERM

Assume T_{r+1} in numerically greatest

Now : $|T_r| \leq |T_{r+1}| \geq |T_{r+2}|$

for expansion $(a + b)^n$ **: Solve :**

$$\frac{(n+1)}{1+\left|\frac{a}{b}\right|} - 1 \leq r \leq \frac{(n+1)}{1+\left|\frac{a}{b}\right|}$$

-1

Question

$\left(1+\frac{4}{3}\right)^8 =$ _____

If $x = \frac{1}{3}$, then the numerically greatest term in the expansion of $(1+4x)^8$ is

A $^8C_4\left(\frac{4}{3}\right)^4$

B $^8C_5\left(\frac{4}{3}\right)^5$

C $^8C_3\left(\frac{4}{3}\right)^3$

D $^8C_6\left(\frac{4}{3}\right)^6$

$(1+4x)^8$ → $(a+b)^n$

$a = 1$

$b = 4x = \frac{4}{3}$

$T_{5+1} = {}^8C_5\left(\frac{4}{3}\right)^5$

$r \le \frac{n+1}{1+\left|\frac{a}{b}\right|}$

$r \le \frac{8+1}{1+\left|\frac{3}{4}\right|}$

$4.14 \le r \le 5.14$

$r = 5$

Question

If $x = 1$, then the numerically greatest term in the expansion of $(3 - 2x)^9$ is

A T_2 & T_3

B T_4

C T_4 & T_5

D None of these

$$\leq r \leq \frac{9+1}{1+\left|\frac{3}{-2}\right|}$$

$$3 \leq r \leq 4$$

$r = 3, 4.$

T_4 T_5.

$(a+b)^n$

$a = 3$

$b = -2x = -2 \times 1 = -2$

$n = 9.$

Question [JEE (Adv.)-2025]

Let $a_0, a_1, \ldots, a_{23}$ be real numbers such that

$$\left(1+\frac{2}{5}x\right)^{23} = \sum_{i=0}^{23} a_i x^i$$

for every real number x. Let a_r be the largest among the numbers a_j for $0 \leq j \leq 23$.

Then the value of r is _____.

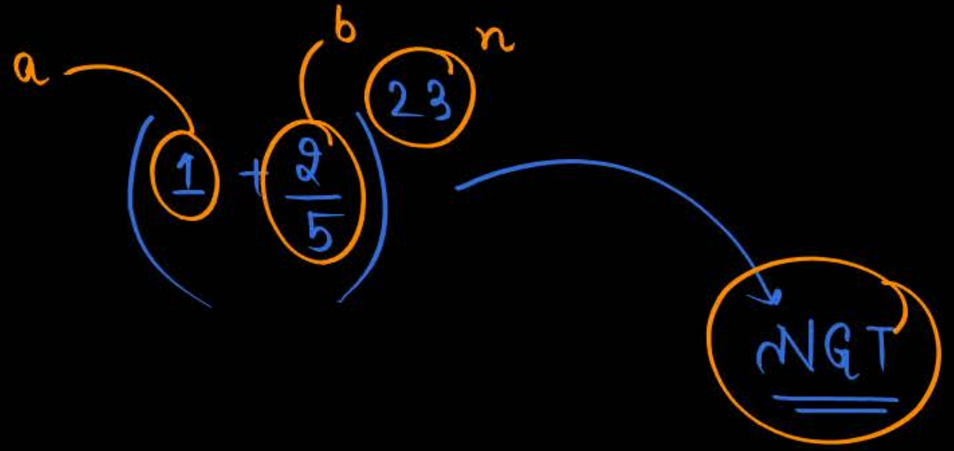
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for

$$\left(1+\frac{2}{5}x\right)^{23} = a_0 + a_1x^1 + a_2x^2 + a_3x^3 + \ldots + a_{23}x^{23}$$

$$a_n = {}^{23}C_n \left(\frac{2}{5}\right)^n$$

$a_0 = {}^{23}C_0$, $a_1x^1 = {}^{23}C_1\frac{2}{5}$, $a_2x^2 = {}^{23}C_2\left(\frac{2}{5}\right)^2$, $a_3x^3 = {}^{23}C_3\left(\frac{2}{5}\right)^3$



$$r \leq \frac{23+1}{1+\frac{5}{2}}$$

$$T_7 = {}^{23}C_6 \left(\frac{2}{5}\right)^6$$

a_6.

$$5.\overline{mm} \leq r \leq 6.\overline{mm}$$

$r = 6$

T_7 is NGT

Question [JEE Mains-2022]

Let for the 9th term in the binomial expansion of $(3 + 6x)^n$, in the increasing powers of $6x$, to be the greatest for $x = \frac{3}{2}$, the least value of n is n_0. If k is the ratio of the coefficient of x^6 to the coefficient of x^3, then $k + n_0$ is equal to



PROPERTIES OF BINOMIAL COEFFICIENT

Binomial coefficient or combinatorial coefficient denoted as

$${}^{n}C_{r} \text{ or } \binom{n}{r} \text{ or } C_{r}$$

Property – 01

(i) $${}^{n}C_{r} = {}^{n}C_{n-r}$$

ex :- ${}^{5}C_{3} = {}^{5}C_{2}$ ✓

${}^{15}C_{7} = {}^{15}C_{8}$ ✓

(ii) If $${}^{n}C_{x} = {}^{n}C_{y}$$ then:

- $x = y$
- OR
- $x + y = n$

Question

If ${}^{23}C_{4n-3} = {}^{23}C_{n+6}$, then sum of all possible values of n is ____7____

$4n-3 = n+6$ OR $4n-3+n+6 = 23$

$3n = 9$

$n = 3$

$5n = 20$

$n = 4$

PROPERTIES OF BINOMIAL COEFFICIENT

Property – 02 : Addition of Binomial coefficient

$$^{n}C_{r} + {}^{n}C_{r-1} = {}^{n+1}C_{r}$$

Proof: $RHS = \frac{n!}{r!(n-r)!} + \frac{n!}{(r-1)!(n-r+1)!}$

$r! = r \cdot (r-1)!$

$(n-r+1)! = (n-r+1) \cdot (n-r)!$

$= \frac{n!}{(r-1)!(n-r)!}\left\{\frac{1}{r} + \frac{1}{n-r+1}\right\}$

$= \frac{n! \cdot (n+1)}{r \cdot (r-1)!\,(n-r+1)(n-r)!} = \frac{(n+1)!}{r!(n-r+1)!}$

RAPID FIRE ROUND

1. ${}^{11}C_8 + {}^{11}C_9 = {}^{12}C_9$.

2. ${}^{22}C_{17} + {}^{22}C_{18} = {}^{23}C_{18}$.

3. ${}^{11}C_8 + {}^{11}C_2 = {}^{12}C_3 = {}^{12}C_9$.

4. ${}^{7}C_0 + {}^{8}C_1 + {}^{9}C_2 + {}^{10}C_3 + {}^{11}C_4 + {}^{12}C_5 = {}^{13}C_5$.

5. ${}^{14}C_{14} + {}^{15}C_{14} + {}^{16}C_{14} + {}^{17}C_{14} + {}^{18}C_{14} = {}^{19}C_{15}$.

④ ${}^{7}C_7 + {}^{8}C_7 + {}^{9}C_7 + {}^{10}C_7 + {}^{11}C_7 + {}^{12}C_7$

$= {}^{13}C_8$

③ ${}^{11}C_8 + {}^{11}C_2$

${}^{11}C_3 + {}^{11}C_2 = {}^{12}C_3$.

1

${}^{7}C_0 + {}^{8}C_1 + {}^{9}C_2 + {}^{10}C_3 + {}^{11}C_4 + {}^{12}C_5 = {}^{13}C_5$.

1

${}^{8}C_0 + {}^{8}C_1$

${}^{9}C_1 + {}^{9}C_2 = {}^{10}C_2 + {}^{10}C_3 = {}^{11}C_3 + {}^{11}C_4 = {}^{12}C_4 + {}^{12}C_5$

⑤ ${}^{14}C_{14} + {}^{15}C_{14} + {}^{16}C_{14} + {}^{17}C_{14} + {}^{18}C_{14} = {}^{19}C_{15}$.

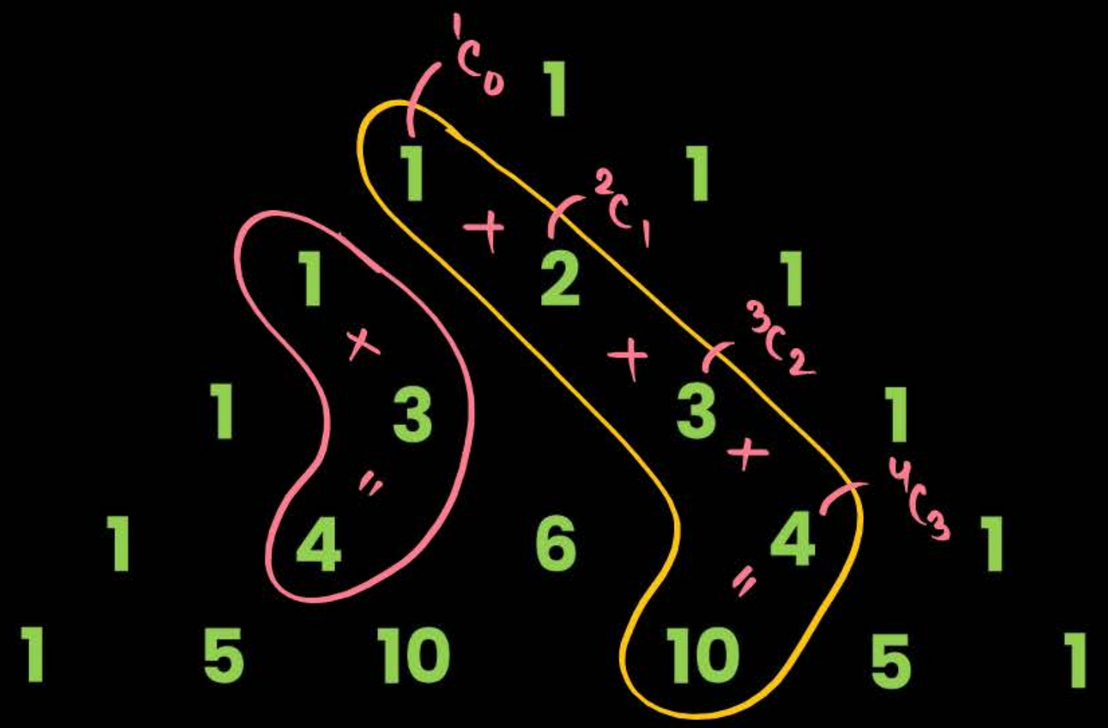
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${}^{15}C_{15} + {}^{15}C_{14}$

${}^{16}C_{15} + {}^{16}C_{14} = {}^{17}C_{15} + {}^{17}C_{14}$

$= {}^{18}C_{15} + {}^{18}C_{14}$

PASCAL'S TRIANGLE



$$^1C_0 + {}^2C_1 + {}^3C_2 + {}^4C_3 = {}^5C_3 \checkmark$$

$$^2C_0 + {}^2C_1 \hookrightarrow {}^3C_1 + {}^3C_2 = {}^4C_2 + {}^4C_3$$

PROPERTIES OF BINOMIAL COEFFICIENT

Property – 03 : Hockey-Stick Identity

$$^{m}C_{m} + {}^{m+1}C_{m} + {}^{m+2}C_{m} + \ldots\ldots + {}^{m+k}C_{m} = {}^{m+k+1}C_{m+1}$$

+1 +1

Question [JEE Mains-2020]

The value of $\sum_{r=0}^{20} {}^{50-r}C_6$ is equal to

A $^{50}C_6 - {}^{30}C_6$

B $^{51}C_7 + {}^{30}C_7$

C $^{51}C_7 - {}^{30}C_7$

D $^{50}C_7 - {}^{30}C_7$

$${}^{50}C_6 + {}^{49}C_6 + {}^{48}C_6 + \ldots\ldots\ldots + {}^{30}C_6 + {}^{29}C_6 + {}^{28}C_6 + \ldots\ldots + {}^{6}C_6$$

$$- \left({}^{6}C_6 + {}^{7}C_6 + {}^{8}C_6 + \ldots\ldots + {}^{29}C_6 \right)$$

$$\Rightarrow {}^{51}C_7 - {}^{30}C_7$$

DO IT BY YOURSELF (DIBY)

DIBY-07

If the constant term in the expansion of $\left(\frac{\sqrt[5]{3}}{x}+\frac{2x}{\sqrt[3]{5}}\right)^{12}$, $x \neq 0$ is $\alpha \times 2^8 \times \sqrt[5]{3}$, then 25α is equal to

[JEE Mains-2024]

[Ans. : 693]

DIBY-08

If the term without x in the expansion of $\left(x^{2/3}+\frac{\alpha}{x^3}\right)^{22}$ is 7315, then $|\alpha|$ is equal to

[JEE Mains-2023]

[Ans. : 1]

DIBY-09

If the coefficient of x^4, x^5 and x^6 in the expansion of $(1+x)^n$ are the arithmetic progression then the maximum value of n is

[JEE Mains-2024]

(A) 7 (B) 21 (C) 28 (D) 14

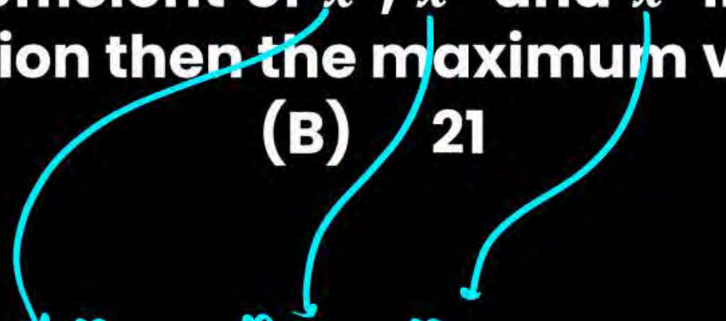
[Ans. : (D)]

DIBY-09

If the coefficient of x^4, x^5 and x^6 in the expansion of $(1 + x)^n$ are the arithmetic progression then the maximum value of n is [JEE Mains-2024]

(A) 7 (B) 21 (C) 28 (D) 14

[Ans. : (D)]



$${}^nC_4, {}^nC_5, {}^nC_6 \rightarrow \text{AP.}$$

$$T_{r+1} = {}^nC_r x^r.$$

$$2 \cdot {}^nC_5 = {}^nC_4 + {}^nC_6.$$

$$2 \cdot \frac{n!}{5!(n-5)!} = \frac{n!}{4!(n-4)!} + \frac{n!}{6!(n-6)!}$$

$$\frac{2}{5(n-5)} = \frac{1}{(n-4)(n-5)} + \frac{1}{30}.$$

DIBY-13

If the coefficient of x^7 in $\left(ax - \frac{1}{bx^2}\right)^{13}$ and the coefficient of x^{-5} in $\left(ax + \frac{1}{bx^2}\right)^{13}$ are equal, then a^4b^4 is equal to [JEE Mains-2023]

(A) 22 (B) 33 (C) 44 (D) 11

[Ans. : (A)]

DIBY-14

If the second, third and fourth terms in the expansion of $(x + y)^n$ are 135, 30 and 10/3, respectively, then $6(n^3 + x^2 + y)$ is equal to [JEE Mains-2024]

[Ans. : 806]

DIBY-15

The sum of the coefficient of $x^{2/3}$ and $x^{-2/5}$ in the binomial expansion of $\left(x^{2/3} + \frac{1}{2}x^{-2/5}\right)^9$ is [JEE Mains-2024]

(A) 21/4 (B) 63/16 (C) 19/4 (D) 69/16

[Ans. : (A)]

DIBY-16

If ${}^nC_4, {}^nC_5$ and ${}^nC_6$ are in A.P., then n can be [JEE Mains-2019]

(A) 9 (B) 14 (C) 11 (D) 12

[Ans. : (B)]

DIBY-17

In the expansion $\left(\frac{x}{\cos\theta}+\frac{1}{x\sin\theta}\right)^{16}$, if l_1 is the least value of the term independent of x when $\frac{\pi}{8} \leq \theta \leq \frac{\pi}{4}$ and l_2 is the least value of the term independent of x when $\frac{\pi}{16} \leq \theta \leq \frac{\pi}{8}$, then the ratio $l_2 : l_1$ is equal to [JEE Mains-2020]

(A) 1 : 8 (B) 1 : 16 (C) 8 : 1 (D) 16 : 1

[Ans. : (D)]

DIBY-18

Let α be the constant term in the binomial expansion of $\left(\sqrt{x}-\frac{6}{x^{3/2}}\right)^n, n \leq 15$. If the sum of the coefficients of the remaining terms in the expansion is 649 and the coefficient of x^{-n} is $\lambda\alpha$, then λ is equal to [JEE Mains-2023]

[Ans. : 36]

DIBY-19

The sum of the coefficient of $x^{2/3}$ and $x^{-2/5}$ in the binomial expansion of $\left(x^{2/3}+\frac{1}{2}x^{-2/5}\right)^{9}$ is [JEE Mains-2024]

(A) $\frac{21}{4}$ (B) $\frac{69}{16}$ (C) $\frac{63}{16}$ (D) $\frac{19}{4}$

[Ans. : (A)]

DIBY-20

Let $0 \leq r \leq n$. If ${}^{n+1}C_{r+1} : {}^{n}C_{r} : {}^{n-1}C_{r-1} = 55 : 35 : 21$, then $2n + 5r$ is equal to [JEE Mains-2024]

(A) 60 (B) 62 (C) 50 (D) 55

[Ans. : (C)]

DIBY-21

For some $n \neq 10$, let the coefficients of the 5th, 6th and 7th terms in the binomial expansion of $(1+x)^{n+4}$ be in A.P. Then the largest coefficient in expansion of $(1+x)^{n+4}$ is [JEE Mains-2025]

(A) 35 (B) 20 (C) 10 (D) 70

[Ans. : (A)]

DIBY-22

A ratio of the 5th term from the beginning to the 5th term from the end in the binomial expansion of $\left(2^{1/3}+\frac{1}{2(3)^{1/3}}\right)^{10}$ is [JEE Mains-2019]

(A) $1:2(6)^{1/3}$ (B) $1:4(16)^{1/3}$ (C) $4(36)^{1/3}:1$ (D) $2(36)^{1/3}:1$

[Ans. : (C)]

DIBY-23

If α and β be the coefficients of x^4 and x^2 respectively in the expansion of $\left(x+\sqrt{x^2-1}\right)^6+\left(x-\sqrt{x^2-1}\right)^6$, then [JEE Mains-2020]

(A) $\alpha+\beta=-30$ (B) $\alpha-\beta=60$ (C) $\alpha-\beta=-132$ (D) $\alpha+\beta=60$

[Ans. : (C)]

DIBY-24

If the maximum value of the term independent of t in the expansion of $\left(t^2x^{1/5}+\frac{(1-x)^{1/10}}{t}\right)^{15}$, $x\geq 0$ is K, then $8K$ is equal to [JEE Mains-2022]

[Ans. : 6006]

DIBY-25

Let the coefficients of x^{-1} and x^{-3} in the expansion of $\left(2x^{1/5} - \frac{1}{x^{1/5}}\right)^{15}$ $x > 0$ be m and n respectively. If r is a positive integer such that $mn^2 = {}^{15}C_r \cdot 2^r$, then the value of r is equal to [JEE Mains-2022]

[Ans. : 5]

DIBY-26

If a, b and c are the greatest values of ${}^{19}C_p$, ${}^{20}C_q$ and ${}^{21}C_r$, respectively, then [JEE Mains-2020]

(A) $\frac{a}{11} = \frac{b}{11} = \frac{c}{42}$ (B) $\frac{a}{11} = \frac{b}{11} = \frac{c}{42}$ (C) $\frac{a}{11} = \frac{b}{11} = \frac{c}{42}$ (D) $\frac{a}{11} = \frac{b}{11} = \frac{c}{42}$

[Ans. : (C)]

DIBY-27

Let the ratio of the fifth term from the beginning to the fifth term from the end in the binomial expansion of $\left(\sqrt[4]{2} + \frac{1}{\sqrt[4]{3}}\right)^n$, in the increasing powers of $\frac{1}{\sqrt[4]{3}}$ be $\sqrt[4]{6}$: 1. If the sixth term from the beginning is $\frac{\alpha}{\sqrt[4]{3}}$, then α is equal to [JEE Mains-2022]

[Ans. : 84]



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians